

Dimensionless Scaling and Physical Non-Equivalence Between Atomic and Planetary Systems: A Quantitative Comparison of Gold-197 and the Sun–Earth System

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Abstract

The apparent structural resemblance between atomic and planetary systems has played an important role in the historical development and conceptual interpretation of atomic models. Although the planetary picture provides an intuitive representation of a compact central object surrounded by smaller constituents, modern atomic physics demonstrates that the underlying physical descriptions of atomic and planetary systems are fundamentally different. In this work, we develop a quantitative dimensionless-scaling framework to examine the limits of the atomic–planetary analogy through a comparison between a gold-197 atomic system and the Sun–Earth system. Characteristic length, velocity, time, mass, energy, force, and angular-momentum scales are evaluated under explicitly stated physical assumptions and subsequently expressed through normalized ratios. The atomic quantities derived from the classical model are treated as characteristic scales rather than literal electron trajectories or quantum observables. For the initial Gold-197–Sun–Earth comparison, the normalized characteristic length and velocity ratios are approximately 95.94 and 395.82, respectively, while the corresponding mass and energy ratios are approximately 1.0864 and 2.39×10^{-50} . These results demonstrate that an apparent structural resemblance does not imply comparable physical scales or dynamical equivalence. The analysis is subsequently extended through sensitivity studies and comparisons across different characteristic atomic and planetary scales in order to determine the robustness of the observed relationships. The resulting framework provides a quantitative approach for distinguishing the positive structural aspects of the atomic–planetary analogy from the large-scale physical differences imposed by electromagnetic interactions, gravitation, and quantum mechanics.

Keywords: atomic systems; planetary systems; dimensionless scaling; Gold-197; Sun–Earth system; atomic–planetary analogy; quantum mechanics; electromagnetic interaction; gravitational interaction

1. Introduction

The analogy between atomic and planetary systems has played an important role in the historical development of atomic theory. In the early twentieth century, the planetary picture provided a useful conceptual framework for describing a positively charged central nucleus surrounded by electrons. Bohr’s 1913 model incorporated this picture into an early quantum description of the atom and used classical mechanical concepts together with quantum conditions to account for the observed hydrogen spectrum [1, 2, 3]. Subsequent historical studies have emphasized that the development of the old quantum theory was closely connected with mathematical ideas originating in celestial mechanics [4, 5, 6]. The analogy between planetary motion and atomic motion therefore has both historical and conceptual significance.

Despite this structural similarity, atomic and planetary systems are not physically equivalent. Planetary motion is predominantly governed by gravitation and can often be described to a high degree of accuracy using classical

mechanics. Atomic systems, in contrast, require quantum mechanics for their fundamental description, while electromagnetic interactions determine their characteristic electronic structure. Consequently, an electron in an atom cannot be regarded as a classical particle following a definite planetary orbit. The historical development of the Bohr model itself illustrates this distinction: similarities between planetary and atomic systems provided useful guidance, but important differences between the two systems were essential to the development of quantum theory [4, 7]. Thus, a structural correspondence between two physical systems does not by itself imply a dynamical or physical equivalence.

The distinction between structural analogy and physical equivalence raises a quantitative question. If corresponding characteristic scales are constructed for an atomic system and a planetary system, can their similarities and differences be expressed independently of the units used to describe them? Dimensionless quantities provide a natural framework for addressing this ques-

tion because they permit comparisons between systems whose absolute length, mass, velocity, and time scales differ by many orders of magnitude. Such a framework can also reveal whether an apparent correspondence observed in one physical quantity is accompanied by comparable behavior in other quantities.

In the present study, we develop a dimensionless characteristic-scale framework for a direct comparison between a Gold-197 atomic system and the Sun–Earth system. Gold-197 is selected as a reference atomic system because its nuclear charge and mass number provide explicitly defined parameters for constructing characteristic nuclear and atomic scales. The Sun–Earth system provides a corresponding planetary reference characterized by the solar mass, Earth mass, Earth–Sun separation, and Earth’s orbital motion. Within the adopted model, characteristic length, velocity, timescale, mass hierarchy, interaction energy, interaction force, and angular momentum are constructed for both systems.

A central feature of the present analysis is that the atomic quantities obtained from the classical Coulomb-orbit construction are not interpreted as exact quantum-mechanical observables. They are used only as characteristic scales within a clearly defined mathematical model. This distinction is particularly important for Gold-197, which is a many-electron quantum system and cannot be represented literally by a single classical electron orbit. The purpose of the classical construction is therefore to establish a consistent scale for comparison rather than to propose a classical replacement for quantum atomic theory.

The main research gap addressed here is the lack of a unified multi-quantity comparison in which the atomic–planetary analogy is tested simultaneously across several dimensionless characteristic scales. A correspondence in one quantity, such as a mass hierarchy, does not necessarily imply correspondence in length, velocity, timescale, energy, force, or angular momentum. Moreover, the numerical values of such ratios can depend on the characteristic scales adopted for the atomic system. A meaningful assessment therefore requires not only the calculation of the dimensionless ratios but also an examination of their sensitivity to the underlying model parameters.

Accordingly, the present work has two primary objectives. First, we construct and evaluate dimensionless ratios for the seven characteristic quantities listed above for the Gold-197–Sun–Earth comparison. Second, we investigate the robustness of these ratios by varying the characteristic atomic length, nuclear charge, and mass number. This allows us to distinguish numerical correspondences that persist under changes in the model parameters from those that arise primarily from a particular choice of characteristic scale.

The central research question is therefore whether a dimensionless characteristic-scale framework can provide a quantitative criterion for distinguishing structural similarity from physical equivalence between atomic and planetary systems. Rather than searching for a single universal scaling factor, we examine each physical quantity independently and determine the degree of scale correspondence or separation that it exhibits. The resulting framework is intended to provide a reproducible quantitative description of the atomic–planetary analogy while explicitly retaining the physical limitations of the underlying classical and two-body approximations.

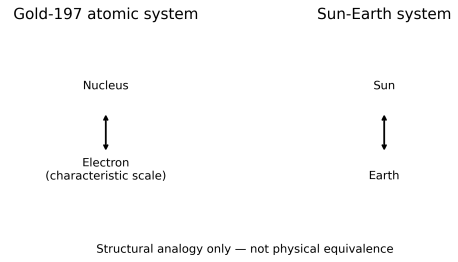


Figure 1: Schematic representation of the structural analogy between the Gold–197 atomic system and the Sun–Earth system. The comparison illustrates structural correspondence only and does not imply physical equivalence

2. Physical Systems and Model Assumptions

The comparison is constructed between two deliberately simplified reference systems: a Gold-197 atomic system and the Sun–Earth planetary system. The purpose of this section is to define the physical parameters and characteristic scales used in the subsequent dimensionless analysis. The adopted quantities are reference scales within the model and are not intended to constitute complete microscopic or dynamical descriptions of the corresponding physical systems.

2.1. Gold-197 Atomic System

The atomic reference system is neutral Gold-197, with atomic number $Z = 79$ and mass number $A = 197$. The atomic number determines the nuclear charge, while the mass number specifies the number of nucleons in the nucleus. Gold atomic data and the corresponding electronic structure are documented in the NIST Atomic Spectra Database [8, 2].

For the nuclear scale, we use the standard empirical nuclear-radius parameterization

$$R_N = r_0 A^{1/3}, \quad (1)$$

where $r_0 = 1.2 \text{ fm}$ is adopted as the characteristic

nuclear-radius coefficient [3, 9, 10]. For $A = 197$, this gives

$$R_N \approx 6.982 \text{ fm}. \quad (2)$$

The nuclear radius obtained from this expression is used only as a characteristic nuclear length. It should not be interpreted as an exact experimental charge radius of the Gold-197 nucleus, since experimentally measured nuclear charge radii depend on the detailed definition of the nuclear radius and on the underlying nuclear structure [9].

A characteristic atomic length of

$$r_{\text{Au}} = 0.144 \text{ nm} \quad (3)$$

is adopted as the reference atomic scale. This quantity is treated as a characteristic electronic length rather than as the radius of a literal classical electron orbit. The choice is therefore regarded as a model parameter, and its effect on the dimensionless comparisons is examined explicitly in the sensitivity analysis of Sec. 6.1.

For the purpose of constructing a characteristic electromagnetic scale, the nucleus–electron interaction is represented by a central Coulomb form,

$$F_C = \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r_{\text{Au}}^2}. \quad (4)$$

This expression is not intended to represent the exact force experienced by an electron in neutral Gold. In a real many-electron atom, electron screening, electron–electron interactions, and relativistic effects modify the electronic potential. Instead, F_C defines a controlled reference electromagnetic scale associated with the nuclear charge Z .

Using this reference force scale, a classical characteristic velocity is defined through the circular-motion condition

$$\frac{m_e v_{\text{Au}}^2}{r_{\text{Au}}} = \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r_{\text{Au}}^2}, \quad (5)$$

which gives

$$v_{\text{Au}} = \sqrt{\frac{1}{4\pi\epsilon_0} \frac{Ze^2}{m_e r_{\text{Au}}}}. \quad (6)$$

The corresponding characteristic timescale is then defined by

$$T_{\text{Au}} = \frac{2\pi r_{\text{Au}}}{v_{\text{Au}}}. \quad (7)$$

The quantities v_{Au} and T_{Au} are classical characteristic scales introduced for normalization. They are not interpreted as the actual velocity or orbital period of an electron in Gold-197. This distinction is essential because the electronic structure of an atom requires a quantum-mechanical description [8, 6].

2.2. Sun–Earth System

The planetary reference system consists of the Sun and Earth. The solar mass and Earth mass define the characteristic planetary mass hierarchy, while the mean Earth–Sun separation provides the characteristic planetary length scale. The corresponding astronomical parameters are taken from standard astronomical data sources [11, 12].

The gravitational interaction is represented by the Newtonian two-body force,

$$F_G = G \frac{M_\odot M_\oplus}{r_{\text{SE}}^2}, \quad (8)$$

where $M_\odot$ is the solar mass, $M_\oplus$ is the Earth mass, and r_{SE} is the mean Earth–Sun separation.

For a circular two-body orbit, the corresponding characteristic velocity is

$$v_\oplus = \sqrt{\frac{GM_\odot}{r_{\text{SE}}}}. \quad (9)$$

The measured characteristic orbital velocity of Earth is used in the numerical comparison, while the above expression provides the corresponding idealized dynamical relation. The two-body approximation is adopted because the purpose of the analysis is to construct a well-defined planetary reference scale rather than to model the complete Solar System dynamics [13].

The corresponding characteristic planetary timescale is defined as

$$T_\oplus = \frac{2\pi r_{\text{SE}}}{v_\oplus}. \quad (10)$$

2.3. Model Assumptions

The two systems differ fundamentally in their underlying physics. Consequently, the comparison is performed at the level of characteristic scales rather than by assuming that the two systems obey identical dynamical equations.

The principal assumptions are as follows. First, the atomic system is represented by a central Coulomb reference scale rather than by a complete many-electron quantum-mechanical calculation. Second, the adopted value of r_{Au} is treated as a characteristic atomic length and not as a unique electron-orbit radius. Third, the nuclear radius is described by the empirical $r_0 A^{1/3}$ parameterization rather than by a detailed nuclear-structure model. Fourth, the planetary system is treated as an idealized Sun–Earth two-body system. Finally, all comparisons are normalized into dimensionless ratios before physical interpretation.

These assumptions deliberately separate the mathematical comparison from claims of dynamical equivalence. The framework is therefore designed to determine how closely the selected characteristic scales correspond

and how strongly the resulting ratios depend on the specific physical scales adopted.

The atomic mass hierarchy is characterized by the ratio between the nuclear mass and the electron mass,

$$\mu_{\text{Au}} \frac{M_N}{m_e}. \quad (11)$$

For the planetary system, the corresponding hierarchy is

$$\mu_{\text{SE}} \frac{M_{\odot}}{M_{\oplus}}. \quad (12)$$

The normalized mass-hierarchy ratio is therefore

$$\mathcal{R}_M \frac{\mu_{\text{Au}}}{\mu_{\text{SE}}}. \quad (13)$$

For the systems considered here,

$$\mu_{\text{Au}} \approx 3.6172 \times 10^5, \quad (14)$$

while

$$\mu_{\text{SE}} \approx 3.3294 \times 10^5. \quad (15)$$

Consequently,

$$\mathcal{R}_M \approx 1.0864. \quad (16)$$

The proximity of this ratio to unity indicates that the central-to-orbiting-body mass hierarchy is numerically similar under the definitions adopted here. This result, however, does not imply dynamical equivalence because the interactions and governing theories of the two systems are different.

2.4. Characteristic Energy Ratio

The magnitude of the Coulomb interaction energy at the adopted characteristic atomic scale is defined as

$$E_{\text{Au}} \frac{k_e Z e^2}{r_{\text{Au}}}. \quad (17)$$

For the Sun–Earth system, the corresponding gravitational interaction-energy scale is

$$E_{\text{SE}} \frac{G M_{\odot} M_{\oplus}}{r_{\text{SE}}}. \quad (18)$$

The dimensionless energy comparison is

$$\mathcal{R}_E \frac{E_{\text{Au}}}{E_{\text{SE}}}. \quad (19)$$

The computational results give

$$E_{\text{Au}} \approx 1.266 \times 10^{-16} \text{ J}, \quad (20)$$

and

$$E_{\text{SE}} \approx 5.298 \times 10^{33} \text{ J}, \quad (21)$$

leading to

$$\mathcal{R}_E \approx 2.39 \times 10^{-50}. \quad (22)$$

The atomic quantity in Eq. (17) is referred to as a characteristic Coulomb energy scale rather than a precise atomic binding energy.

2.5. Interaction-Force Ratio

The characteristic electromagnetic and gravitational forces are defined by

$$F_{\text{EM}} \frac{k_e Z e^2}{r_{\text{Au}}^2} \quad (23)$$

and

$$F_{\text{G}} \frac{G M_{\odot} M_{\oplus}}{r_{\text{SE}}^2}, \quad (24)$$

respectively.

Their direct ratio is

$$\mathcal{R}_F \frac{F_{\text{EM}}}{F_{\text{G}}}. \quad (25)$$

For the selected characteristic scales, the computational analysis gives

$$F_{\text{EM}} \approx 8.79 \times 10^{-7} \text{ N}, \quad (26)$$

and

$$F_{\text{G}} \approx 3.54 \times 10^{22} \text{ N}, \quad (27)$$

resulting in

$$\mathcal{R}_F \approx 2.48 \times 10^{-29}. \quad (28)$$

This ratio is interpreted only as a comparison of the two selected interaction-force scales. It is not treated as a universal comparison of the intrinsic strengths of electromagnetism and gravitation.

2.6. Angular-Momentum Scale

3. Mathematical Framework and Dimensionless Scaling

The comparison between the Gold-197 atomic system and the Sun–Earth system is formulated in terms of dimensionless characteristic-scale ratios. The objective is not to construct a transformation that maps one physical system onto the other, but to quantify the relative separation between corresponding characteristic scales.

Because the physical meaning of a characteristic quantity differs between the two systems, a single universal normalization cannot be imposed on every observable. Instead, each dimensionless ratio is defined from a physically corresponding pair of characteristic quantities. This approach ensures that the normalization remains explicit and reproducible for every comparison.

The framework considers seven characteristic quantities: length, velocity, timescale, mass hierarchy, interaction energy, interaction force, and angular momentum. All dimensional quantities are calculated first in SI units, after which the corresponding dimensionless ratios are constructed.

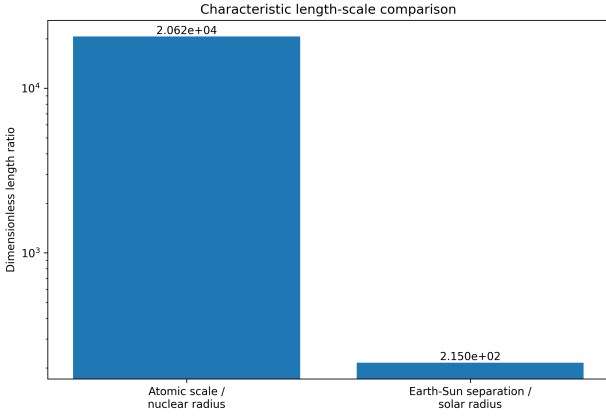


Figure 2: Dimensionless scaling framework used to compare the Gold-197 atomic system with the Sun–Earth system. The framework maps corresponding characteristic quantities and defines the dimensionless ratios used throughout the analysis. The comparison represents characteristic-scale correspondence only and does not imply physical equivalence between the two systems.

3.1. Length-Scale Ratio

For the atomic system, the dimensionless internal length scale is defined as the ratio of the characteristic atomic length to the nuclear radius,

$$\Lambda_{\text{Au}} = \frac{r_{\text{Au}}}{R_N}. \quad (29)$$

For the planetary system, the corresponding internal length scale is defined as the ratio of the Earth–Sun separation to the solar radius,

$$\Lambda_{\text{SE}} = \frac{r_{\text{SE}}}{R_{\odot}}. \quad (30)$$

The normalized length-scale comparison is therefore defined as

$$R_L = \frac{\Lambda_{\text{Au}}}{\Lambda_{\text{SE}}} = \frac{r_{\text{Au}}/R_N}{r_{\text{SE}}/R_{\odot}}. \quad (31)$$

This definition compares the relative extent of the characteristic atomic and planetary length scales with respect to their respective central-body radii.

3.2. Velocity-Scale Ratio

The atomic characteristic velocity is defined by the classical central-force construction introduced in Sec. 2.1,

$$v_{\text{Au}} = \sqrt{\frac{k_e Z e^2}{m_e r_{\text{Au}}}}, \quad (32)$$

where

$$k_e = \frac{1}{4\pi\epsilon_0}. \quad (33)$$

The corresponding planetary characteristic velocity is the orbital velocity of Earth around the Sun,

$$v_{\oplus} = \sqrt{\frac{GM_{\odot}}{r_{\text{SE}}}}. \quad (34)$$

The dimensionless velocity-scale ratio is then

$$R_v = \frac{v_{\text{Au}}}{v_{\oplus}}. \quad (35)$$

The quantity R_v therefore measures the separation between the two adopted characteristic velocity scales. It does not imply that an electron in Gold-197 follows a classical orbit with velocity v_{Au} .

3.3. Timescale Ratio

The characteristic atomic timescale is defined from the adopted atomic length and characteristic velocity,

$$T_{\text{Au}} = \frac{2\pi r_{\text{Au}}}{v_{\text{Au}}}. \quad (36)$$

For the planetary system, the corresponding characteristic timescale is

$$T_{\oplus} = \frac{2\pi r_{\text{SE}}}{v_{\oplus}}. \quad (37)$$

The dimensionless timescale ratio is

$$R_T = \frac{T_{\text{Au}}}{T_{\oplus}}. \quad (38)$$

A value substantially smaller than unity indicates that the characteristic timescale defined by the atomic model is much shorter than the corresponding planetary timescale.

3.4. Mass-Hierarchy Ratio

The atomic mass hierarchy is defined by the ratio of the nuclear mass to the electron mass,

$$\mu_{\text{Au}} = \frac{M_N}{m_e}. \quad (39)$$

For the planetary system, the corresponding central-to-orbiting-body mass hierarchy is

$$\mu_{\text{SE}} = \frac{M_{\odot}}{M_{\oplus}}. \quad (40)$$

The normalized mass-hierarchy ratio is therefore

$$R_M = \frac{\mu_{\text{Au}}}{\mu_{\text{SE}}} = \frac{M_N/m_e}{M_{\odot}/M_{\oplus}}. \quad (41)$$

This ratio measures whether the relative dominance of the central body over the orbiting body is comparable in the two systems. A value near unity represents numerical similarity of this particular hierarchical quantity only and does not imply similarity of the underlying dynamics.

3.5. Interaction-Energy Ratio

The magnitude of the characteristic Coulomb interaction energy is

$$E_{\text{Au}} = k_e \frac{Ze^2}{r_{\text{Au}}}. \quad (42)$$

For the Sun–Earth system, the corresponding gravitational interaction-energy scale is taken as

$$E_{\text{SE}} = G \frac{M_{\odot} M_{\oplus}}{r_{\text{SE}}}. \quad (43)$$

The dimensionless interaction-energy ratio is

$$R_E = \frac{E_{\text{Au}}}{E_{\text{SE}}}. \quad (44)$$

The atomic quantity in this definition is a characteristic Coulomb energy scale and is not interpreted as an exact electronic binding energy of Gold-197.

3.6. Interaction-Force Ratio

The characteristic electromagnetic force scale is

$$F_{\text{EM}} = k_e \frac{Ze^2}{r_{\text{Au}}^2}, \quad (45)$$

whereas the corresponding gravitational force scale for the Sun–Earth system is

$$F_G = G \frac{M_{\odot} M_{\oplus}}{r_{\text{SE}}^2}. \quad (46)$$

The normalized force-scale ratio is

$$R_F = \frac{F_{\text{EM}}}{F_G}. \quad (47)$$

This ratio compares the two selected characteristic interaction scales. It must not be interpreted as a universal ratio of the fundamental strengths of electromagnetism and gravity.

3.7. Angular-Momentum Scale

Within the classical characteristic-scale construction, the atomic angular-momentum scale is defined as

$$J_{\text{Au}} = m_e v_{\text{Au}} r_{\text{Au}}. \quad (48)$$

The corresponding planetary orbital angular momentum is

$$J_{\oplus} = M_{\oplus} v_{\oplus} r_{\text{SE}}. \quad (49)$$

The dimensionless angular-momentum ratio is therefore

$$R_J = \frac{J_{\text{Au}}}{J_{\oplus}}. \quad (50)$$

The quantity J_{Au} is a classical characteristic scale introduced for comparison. It is not interpreted as a quantum angular-momentum eigenvalue of a Gold electron.

3.8. Interpretation of the Dimensionless Framework

The seven ratios defined above are intentionally independent comparisons rather than components of a single universal scaling factor. Consequently, agreement in one ratio does not require agreement in any of the others.

This distinction is central to the analysis. If a normalized ratio is close to unity, the result indicates numerical similarity of the specific characteristic quantity under the adopted normalization. It does not establish dynamical, structural, or physical equivalence between the atomic and planetary systems.

The resulting framework therefore provides a quantitative measure of scale correspondence while retaining the physical distinctions between the Coulomb-based atomic reference model and the gravity-based planetary reference model. The numerical evaluation of these ratios is presented separately in Sec. 5, while their dependence on the adopted atomic and nuclear parameters is examined in Sec. 6.

4. Computational Method and Reproducibility

The numerical analysis was implemented using a Python-based computational workflow. The purpose of the computation is to evaluate the characteristic quantities defined in Sec. 3, maintain consistent SI units, construct the corresponding dimensionless ratios, and perform the parameter-dependent sensitivity analysis described later in the paper.

All physical constants and system parameters are specified explicitly at the beginning of the calculation. The Gold-197 system is defined using $Z = 79$, $A = 197$, the adopted nuclear-radius coefficient $r_0 = 1.2$ fm, and the characteristic atomic length $r_{\text{Au}} = 0.144$ nm. The nuclear radius is calculated from

$$R_N = r_0 A^{1/3}. \quad (51)$$

The planetary reference system is defined using the solar mass, Earth mass, Earth–Sun separation, solar radius, and Earth’s characteristic orbital velocity.

The computational procedure follows the mathematical definitions introduced in Sec. 3. Dimensional quantities are calculated first, and the dimensionless ratios are constructed only after the corresponding atomic and planetary quantities have been evaluated. This order of calculation minimizes unit inconsistencies and makes the numerical procedure directly traceable to the analytical framework.

The computational workflow consists of the following steps:

1. Define the physical constants and reference parameters for the Gold-197 and Sun–Earth systems.
2. Calculate the Gold-197 nuclear radius using $R_N = r_0 A^{1/3}$.

3. Calculate the atomic characteristic length, velocity, timescale, interaction energy, interaction force, and angular momentum.
4. Calculate the corresponding characteristic quantities for the Sun–Earth system using the adopted gravitational relations.
5. Construct the seven dimensionless ratios R_L , R_v , R_T , R_M , R_E , R_F , and R_J .
6. Store the calculated quantities using consistent SI units and retain sufficient numerical precision for subsequent analysis.
7. Compare the resulting characteristic scales and generate the numerical data used in the results section.
8. Repeat the calculation over selected ranges of r_{Au} , Z , and A to quantify the sensitivity of the dimensionless ratios to the adopted model parameters.

The computational implementation is intended as a numerical realization of the mathematical framework rather than as an independent physical model. In particular, the numerical output does not remove the approximations associated with the classical characteristic-scale construction of the atomic system or the two-body approximation of the planetary system.

For reproducibility, the numerical values of the input parameters, the equations used to calculate the characteristic quantities, and the definitions of all dimensionless ratios are explicitly stated in Secs. 2 and 3. The complete Python implementation can be supplied as supplementary material so that the numerical results and sensitivity analysis can be independently reproduced.

5. Results

The mathematical framework defined in Sec. 3 was applied to the reference Gold-197 atomic system and the Sun–Earth system using the parameters specified in Sec. 2. The results are first reported as characteristic dimensional quantities and are then expressed as dimensionless ratios. This separation allows the numerical output of the model to be distinguished from its subsequent physical interpretation.

5.1. Characteristic Physical Scales

For the Gold-197 reference system, the adopted nuclear-radius parameterization gives

$$R_N \approx 6.982 \text{ fm.} \quad (52)$$

The corresponding dimensionless atomic internal length scale is

$$\frac{r_{Au}}{R_N} \approx 2.0623 \times 10^4. \quad (53)$$

For the Sun–Earth system, the corresponding planetary internal length scale is

$$\frac{r_{SE}}{R_\odot} \approx 2.1503 \times 10^2. \quad (54)$$

Consequently, the normalized length-scale ratio is

$$R_L \approx 9.594 \times 10^1. \quad (55)$$

The characteristic velocity obtained from the adopted classical Coulomb scale is

$$v_{Au} \approx 1.1787 \times 10^7 \text{ m s}^{-1}, \quad (56)$$

while the characteristic Earth orbital velocity is

$$v_\oplus \approx 2.978 \times 10^4 \text{ m s}^{-1}. \quad (57)$$

The resulting normalized velocity ratio is

$$R_v \approx 3.958 \times 10^2. \quad (58)$$

The corresponding characteristic timescales are

$$T_{Au} \approx 7.68 \times 10^{-17} \text{ s}, \quad (59)$$

and

$$T_\oplus \approx 3.1563 \times 10^7 \text{ s}, \quad (60)$$

giving

$$R_T \approx 2.432 \times 10^{-24}. \quad (61)$$

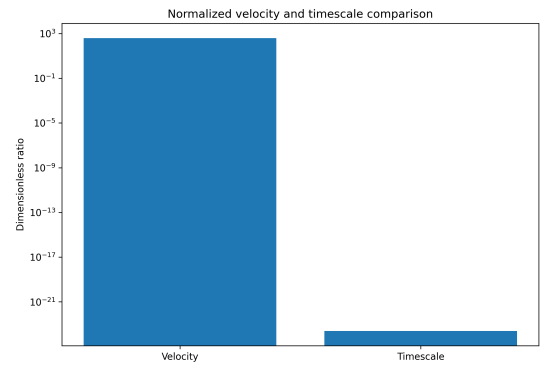


Figure 3: Comparison of the characteristic velocity and timescale scales for the Gold-197 atomic and Sun–Earth systems.

For the mass hierarchy, the atomic and planetary values are

$$\frac{M_N}{m_e} \approx 3.6172 \times 10^5, \quad (62)$$

and

$$\frac{M_{\odot}}{M_{\oplus}} \approx 3.3294 \times 10^5, \quad (63)$$

respectively. Their normalized ratio is therefore

$$R_M \approx 1.0864. \quad (64)$$

The characteristic Coulomb energy scale of the atomic reference system is approximately

$$E_{\text{Au}} \approx 1.266 \times 10^{-16} \text{ J}, \quad (65)$$

whereas the corresponding Sun–Earth gravitational energy scale is

$$E_{\text{SE}} \approx 5.298 \times 10^{33} \text{ J}. \quad (66)$$

Thus,

$$R_E \approx 2.389 \times 10^{-50}. \quad (67)$$

The corresponding characteristic electromagnetic and gravitational force scales are

$$F_{\text{EM}} \approx 8.79 \times 10^{-7} \text{ N}, \quad (68)$$

and

$$F_G \approx 3.54 \times 10^{22} \text{ N}, \quad (69)$$

respectively. The normalized force ratio is therefore

$$R_F \approx 2.482 \times 10^{-29}. \quad (70)$$

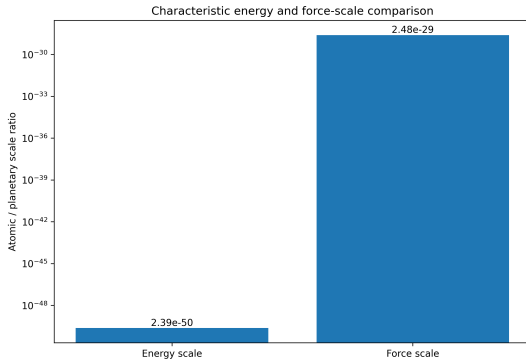


Figure 4: Comparison of the characteristic interaction-energy and force scales of the Gold-197 atomic and Sun–Earth systems. The ratios are shown on a logarithmic scale because of their large separation from unity.

Finally, the classical characteristic angular-momentum scales are

$$J_{\text{Au}} \approx 1.55 \times 10^{-33} \text{ kg m}^2 \text{ s}^{-1}, \quad (71)$$

and

$$J_{\oplus} \approx 2.66 \times 10^{40} \text{ kg m}^2 \text{ s}^{-1}, \quad (72)$$

which gives

$$R_J \approx 5.812 \times 10^{-74}. \quad (73)$$

5.2. Principal Dimensionless Comparisons

The complete set of dimensionless results is summarized in Table 1. The ratios span an exceptionally wide range. The mass hierarchy is the only comparison that remains close to unity, whereas the timescale, energy, force, and angular-momentum ratios are separated from unity by many orders of magnitude.

Table 1: Principal dimensionless characteristic-scale comparisons between the Gold-197 atomic system and the Sun–Earth system.

Quantity	Normalized ratio	Interpretation
Length scale	9.594×10^1	Characteristic-scale comparison
Velocity scale	3.958×10^2	Classical characteristic-velocity comparison
Timescale	2.432×10^{-24}	Characteristic-timescale comparison
Mass hierarchy	1.0864	Central-to-orbiting-body hierarchy
Energy scale	2.389×10^{-50}	Characteristic interaction-energy comparison
Force scale	2.482×10^{-29}	Selected interaction-force comparison
Angular momentum	5.812×10^{-74}	Classical characteristic-scale comparison

The results show that the degree of numerical correspondence depends strongly on the physical quantity being compared. In particular, R_M is close to unity, indicating that the central-to-orbiting-body mass hierarchies are numerically comparable under the adopted definitions. In contrast, the extremely small values of R_T , R_E , R_F , and R_J demonstrate that this similarity does not extend to the corresponding characteristic dynamical scales.

The length and velocity ratios are greater than unity, with values of approximately 95.94 and 395.82, respectively. These values indicate that the adopted atomic characteristic scales are not related to the planetary scales by a common near-unity factor. The very small timescale ratio further indicates the strong separation between the characteristic temporal scales of the two systems.

The energy and force ratios require particular care in interpretation. They compare the specific Coulomb and gravitational characteristic scales defined in Sec. 3; they are not universal measures of the relative fundamental strengths of electromagnetism and gravity. Likewise, the angular-momentum ratio is a comparison of classical characteristic scales and should not be interpreted as

a comparison of quantum angular-momentum observables.

Overall, the reference calculation demonstrates that a near-unity correspondence in one dimensionless property does not imply a general physical or dynamical equivalence between the atomic and planetary systems. The robustness of this conclusion with respect to the adopted atomic and nuclear parameters is examined in Sec. 6.

6. Sensitivity Analysis

The reference results presented in Sec. 5 are obtained for a specific choice of the characteristic atomic length, nuclear charge, and mass number. Because the dimensionless ratios depend on these adopted parameters, a sensitivity analysis is required to determine whether the principal conclusions are robust or arise primarily from a particular reference choice.

Three parameters are varied independently: the characteristic atomic length r_{Au} , the nuclear charge Z , and the mass number A . Unless otherwise stated, the remaining parameters are kept fixed at their reference values, $r_{\text{Au}} = 0.144 \text{ nm}$, $Z = 79$, and $A = 197$.

The parameter dependence can also be obtained analytically from the definitions in Sec. 3. With the planetary quantities held fixed, the principal scaling relations are

$$R_L \propto r_{\text{Au}} A^{-1/3}, \quad (74)$$

$$R_v \propto Z^{1/2} r_{\text{Au}}^{-1/2}, \quad (75)$$

$$R_T \propto Z^{-1/2} r_{\text{Au}}^{3/2}, \quad (76)$$

$$R_M \propto A, \quad (77)$$

$$R_E \propto Z r_{\text{Au}}^{-1}, \quad (78)$$

$$R_F \propto Z r_{\text{Au}}^{-2}, \quad (79)$$

$$R_J \propto Z^{1/2} r_{\text{Au}}^{1/2}. \quad (80)$$

These relations provide an analytical basis for interpreting the numerical sensitivity calculations below.

6.1. Sensitivity to the Characteristic Atomic Length

The characteristic atomic length was varied from 0.050 nm to 0.300 nm while keeping $Z = 79$ and $A = 197$ fixed. The resulting dimensionless ratios are listed in Table 2.

The analytical relations show that increasing r_{Au} increases the normalized length ratio linearly, while the characteristic velocity decreases as $r_{\text{Au}}^{-1/2}$. The characteristic timescale increases as $r_{\text{Au}}^{3/2}$, whereas the characteristic energy and force ratios decrease as r_{Au}^{-1} and r_{Au}^{-2} , respectively.

Table 2: Sensitivity of selected dimensionless characteristic ratios to the characteristic atomic length for $Z = 79$ and $A = 197$.

r_{Au} (nm)	R_L	R_v	R_T	R_E
0.050	3.3301×10^1	6.7172×10^2	4.9757×10^{-25}	6.8802×10^{-50}
0.080	5.3282×10^1	5.3104×10^2	1.0707×10^{-24}	4.3001×10^{-50}
0.100	6.6663×10^1	4.7498×10^2	1.4073×10^{-24}	3.4401×10^{-50}
0.120	7.9924×10^1	4.3360×10^2	1.8500×10^{-24}	2.8667×10^{-50}
0.144	9.5940×10^1	3.9582×10^2	2.4319×10^{-24}	2.3890×10^{-50}
0.160	1.0656×10^2	3.7550×10^2	2.8483×10^{-24}	2.1501×10^{-50}
0.200	1.3321×10^2	3.3586×10^2	3.9806×10^{-24}	1.7200×10^{-50}
0.250	1.6651×10^2	3.0040×10^2	5.5630×10^{-24}	1.3760×10^{-50}
0.300	1.9981×10^2	2.7423×10^2	7.3128×10^{-24}	1.1467×10^{-50}

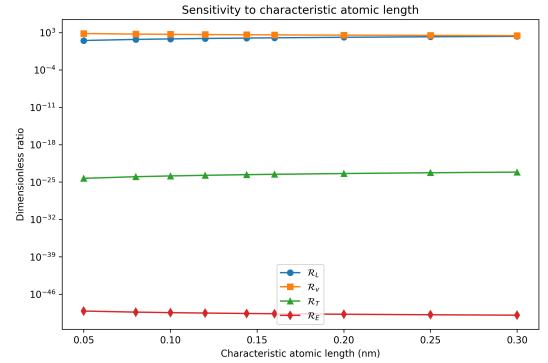


Figure 5: Dependence of the normalized length, velocity, timescale, and energy ratios on the characteristic atomic length. The logarithmic vertical scale is used because the normalized quantities span several orders of magnitude.

The numerical results confirm the analytical scaling relations. Over the explored interval, R_L increases monotonically, whereas R_v decreases. The timescale ratio increases with atomic length, while the energy ratio decreases. Importantly, the qualitative separation between the atomic and planetary characteristic scales remains pronounced throughout the tested range.

6.2. Sensitivity to Nuclear Charge

The dependence on nuclear charge was investigated by varying Z from 10 to 100 while keeping $A = 197$ and $r_{\text{Au}} = 0.144 \text{ nm}$ fixed.

From the analytical scaling relations,

$$R_L \propto A^{-1/3}. \quad (85)$$

$$R_v \propto Z^{1/2}, \quad (81)$$

$$R_E \propto Z, \quad (82)$$

$$R_F \propto Z. \quad (83)$$

The numerical values are summarized in Table 3.

Table 3: Sensitivity of the normalized velocity, energy, and force ratios to nuclear charge for $A = 197$ and $r_{\text{Au}} = 0.144$ nm.

Z	R_v	R_E	R_F
10	1.4083×10^2	3.0240×10^{-51}	3.1415×10^{-30}
20	1.9916×10^2	6.0480×10^{-51}	6.2831×10^{-30}
30	2.4392×10^2	9.0720×10^{-51}	9.4246×10^{-30}
40	2.8155×10^2	1.2096×10^{-50}	1.2566×10^{-29}
50	3.1489×10^2	1.5120×10^{-50}	1.5708×10^{-29}
60	3.4495×10^2	1.8144×10^{-50}	1.8849×10^{-29}
70	3.7259×10^2	2.1168×10^{-50}	2.1991×10^{-29}
80	3.9831×10^2	2.4192×10^{-50}	2.5132×10^{-29}
90	4.2248×10^2	2.7216×10^{-50}	2.8274×10^{-29}
100	4.4533×10^2	3.0240×10^{-50}	3.1415×10^{-29}

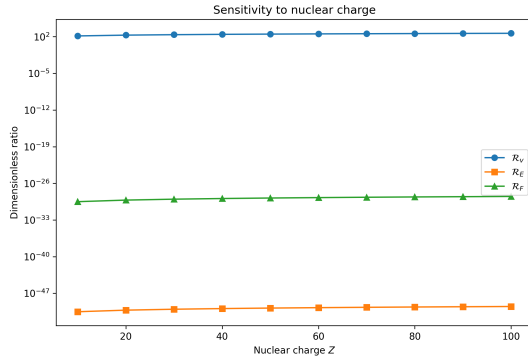


Figure 6: Dependence of the normalized velocity, energy, and force ratios on nuclear charge. The vertical axis is shown on a logarithmic scale because the characteristic ratios span several orders of magnitude.

The numerical results exhibit the expected monotonic behavior. The velocity ratio follows the square-root dependence on Z , whereas the energy and force ratios vary approximately linearly with nuclear charge. Despite these changes, the energy and force ratios remain far below unity throughout the investigated range.

6.3. Sensitivity to Mass Number

The dependence on mass number was investigated by varying A from 50 to 250 while keeping $Z = 79$ and $r_{\text{Au}} = 0.144$ nm fixed.

Because the nuclear radius is modeled as

$$R_N = r_0 A^{1/3}, \quad (84)$$

the normalized length ratio follows

The nuclear mass used in the mass-hierarchy scale is proportional to the mass number within the adopted approximation. Consequently,

$$R_M \propto A. \quad (86)$$

The numerical results are given in Table 4.

Table 4: Sensitivity of the nuclear radius, normalized length ratio, and normalized mass hierarchy to mass number for $Z = 79$ and $r_{\text{Au}} = 0.144$ nm.

A	R_N (fm)	r_{Au}/R_N	R_M
50	4.4208	1.5148×10^2	2.7574×10^{-1}
75	5.0606	1.3233×10^2	4.1362×10^{-1}
100	5.5699	1.2023×10^2	5.5149×10^{-1}
125	6.0000	1.1161×10^2	6.8936×10^{-1}
150	6.3760	1.0503×10^2	8.2723×10^{-1}
175	6.7121	9.9770×10^1	9.6511×10^{-1}
197	6.9824	9.5908×10^1	1.0864
225	7.2986	9.1752×10^1	1.2409
250	7.5595	8.5886×10^1	1.3787

The results demonstrate two distinct trends. Increasing A enlarges the modeled nuclear radius as $A^{1/3}$ and therefore reduces the normalized atomic length scale. At the same time, the central-to-orbiting-body mass hierarchy increases approximately linearly with mass number.

6.4. Combined Interpretation of the Sensitivity Analysis

The three parameter scans demonstrate that the numerical values of the dimensionless ratios depend on the characteristic scales and nuclear parameters adopted in the model. This dependence is expected and does not invalidate the dimensionless comparison; instead, it identifies which conclusions are robust and which are parameter-dependent.

The characteristic length ratio varies systematically with both r_{Au} and A , while the velocity ratio is sensitive to both r_{Au} and Z . The energy and force ratios are particularly sensitive to the characteristic atomic length and nuclear charge because of their inverse-power dependence on r_{Au} and their linear dependence on Z .

In contrast, the qualitative separation between the characteristic timescale, energy, force, and angular-momentum scales remains pronounced over the explored parameter ranges. The mass-hierarchy ratio is the most notable exception: it can approach unity for a restricted range of mass numbers and therefore should not be treated as a universal feature of atomic systems.

The sensitivity analysis therefore supports a more precise interpretation of the main result. The numerical

values of the dimensionless ratios are model-dependent, but the broad separation between most atomic and planetary characteristic scales persists throughout the tested parameter ranges. Consequently, the observed near-unity mass-hierarchy correspondence should be regarded as a specific numerical similarity rather than evidence for a general physical or dynamical equivalence between atomic and planetary systems.

7. Cross-Scale Comparative Interpretation

The results obtained in Secs. 5 and 6 can be interpreted more meaningfully when the seven dimensionless ratios are considered together rather than individually. The purpose of this section is therefore not to introduce additional calculations, but to synthesize the different characteristic-scale comparisons and identify the pattern that emerges across the atomic and planetary systems.

7.1. Hierarchy Versus Dynamical Scales

The most notable feature of the comparison is the difference between the mass hierarchy and the other characteristic quantities. For the reference Gold-197–Sun–Earth comparison, the normalized mass hierarchy is close to unity,

$$R_M \approx 1.0864. \quad (87)$$

This indicates that the ratio between the central-body mass and the orbiting-body mass is numerically comparable in the two selected systems under the adopted definitions.

This correspondence does not, however, extend to the characteristic dynamical scales. The velocity, timescale, energy, force, and angular momentum ratios are substantially displaced from unity. The contrast is particularly pronounced for the timescale, energy, force, and angular-momentum comparisons.

The result demonstrates that similarity in a dimensionless hierarchy does not imply similarity in the complete set of physical scales. Consequently, the atomic–planetary analogy is better regarded as a partial structural correspondence than as a universal scaling relation.

7.2. Near-Unity and Non-Unity Correspondence

A useful distinction can be made between numerical correspondence and physical equivalence. A dimensionless ratio close to unity indicates that two selected characteristic quantities have comparable normalized magnitudes. It does not establish that the corresponding systems possess the same equations of motion, interaction mechanism, or microscopic structure.

The present results provide a clear example of this distinction. The near-unity value of R_M identifies a numerical correspondence in the central-to-orbiting-body

mass hierarchy. In contrast, the large departures of the other ratios from unity demonstrate that this correspondence is not accompanied by a general matching of the characteristic scales.

The length and velocity comparisons also show that the two systems cannot be described by a single scale-independent conversion factor. Their normalized values remain finite and quantitatively comparable, but are not close to unity. The timescale comparison provides an even stronger separation between the two systems.

7.3. Interaction Scales and Physical Interpretation

The energy and force comparisons require particular care because the two systems are governed by different interactions. The atomic reference scale is constructed from a Coulomb interaction, whereas the planetary reference scale is constructed from Newtonian gravitation.

Therefore, the very small values obtained for the normalized energy and force ratios should not be interpreted as universal statements about the fundamental relative strengths of electromagnetism and gravity. They quantify the separation between the specific characteristic scales defined by the present model.

Similarly, the angular-momentum comparison is based on a classical characteristic angular momentum for the atomic reference system. Because the electronic structure of Gold-197 is fundamentally quantum mechanical, this quantity should not be interpreted as a quantum angular-momentum eigenvalue.

7.4. Implications of the Sensitivity Analysis

The sensitivity analysis in Sec. 6 provides an additional test of the interpretation. The numerical values of the dimensionless ratios change systematically when the characteristic atomic length, nuclear charge, or mass number is varied. This demonstrates that the exact numerical values are model-dependent.

Nevertheless, the principal qualitative pattern remains pronounced throughout the explored parameter ranges: the mass hierarchy can approach unity, whereas the characteristic dynamical and interaction scales generally remain strongly separated.

This distinction is important because it prevents the reference Gold-197 result from being interpreted as a universal property of all atomic systems. The observed correspondence is associated with the specific characteristic scales and parameters adopted in the present model.

7.5. Structural Analogy Versus Physical Equivalence

Taken together, the seven characteristic ratios support a clear distinction between structural analogy and physical equivalence. The atomic and planetary systems share a useful mathematical correspondence in the sense that

both can be represented using central-body, orbiting-body, characteristic length, velocity, and timescale concepts. However, the quantitative comparison shows that these correspondences do not produce a common scale transformation across all physical quantities.

The framework therefore supports a restricted interpretation of the atomic–planetary analogy. It can be used to identify specific dimensionless similarities between selected characteristic quantities, but it does not provide evidence for a general dynamical equivalence between atomic and planetary systems.

This cross-scale interpretation provides the basis for the broader discussion of the physical significance, limitations, and possible extensions of the present framework in Sec. 8.

8. Discussion

The quantitative results obtained in this study provide a more detailed assessment of the long-standing analogy between atomic and planetary systems. Historically, the planetary picture played an important role in the development of early atomic theory, particularly in the Bohr model. The connection between celestial mechanics and the old quantum theory has also been examined extensively from a historical and mathematical perspective [1, 4, 5, 14]. The present analysis does not attempt to revive the planetary model as a physical description of the atom. Instead, it uses the analogy as a starting point for a quantitative comparison of characteristic scales.

The most striking result is the contrast between the normalized mass hierarchy and the other characteristic quantities. For the reference Gold-197–Sun–Earth comparison, the normalized mass hierarchy is close to unity,

$$\mathcal{R}_M \approx 1.0864, \quad (88)$$

whereas the normalized energy ratio is approximately

$$\mathcal{R}_E \approx 2.39 \times 10^{-50}. \quad (89)$$

Thus, the relative dominance of the central body over the orbiting body can be numerically similar under the adopted normalization, while the associated interaction-energy scales differ by approximately fifty orders of magnitude. This result demonstrates that a correspondence in one dimensionless quantity cannot be interpreted as evidence for a general scaling equivalence between the two systems.

A similar distinction appears in the length and velocity comparisons. The normalized characteristic length and velocity ratios are approximately 95.91 and 395.82, respectively. These values indicate that the corresponding dimensionless scales are not equal to unity, although they remain finite and can therefore be compared quantitatively. The results should consequently be interpreted

as measures of relative scale separation rather than as scaling constants that transform an atomic system into a planetary system.

The timescale comparison provides an even stronger distinction. The normalized characteristic timescale is approximately

$$\mathcal{R}_T \approx 2.43 \times 10^{-24}. \quad (90)$$

The atomic characteristic process represented by the simplified classical model therefore occurs on a timescale enormously shorter than the orbital timescale of the Earth–Sun system. This large separation is consistent with the fundamentally different characteristic scales associated with atomic and macroscopic dynamical systems.

The force comparison should be interpreted with particular caution. The quantity

$$\mathcal{R}_F = \frac{F_{EM}}{F_G} \approx 2.48 \times 10^{-29} \quad (91)$$

compares the electromagnetic force calculated for the selected Gold-197 characteristic scale with the gravitational force calculated for the selected Sun–Earth scale. It should not be interpreted as a universal dimensionless measure of the intrinsic strength of electromagnetism relative to gravity. The two forces act in different physical systems and are evaluated at very different characteristic length and mass scales.

The same caution applies to the characteristic energy and angular-momentum comparisons. The quantity used for the atomic energy is a Coulomb interaction-energy scale derived from the classical characteristic model. It is not a precision calculation of the binding energy of an electron in gold. Likewise, the quantity $m_e v_{Au} r_{Au}$ is a classical angular-momentum scale and should not be interpreted as the quantum-mechanical orbital angular momentum of a gold electron.

This distinction is essential because the historical planetary analogy belongs to the development of the old quantum theory, whereas modern quantum mechanics replaced the notion of definite classical electron trajectories with a quantum description. Historical studies emphasize both the usefulness of the planetary analogy and the conceptual changes required as quantum mechanics developed [4, 7, 15]. The present calculations therefore deliberately retain the classical construction only as a characteristic-scale model.

The sensitivity analysis strengthens this interpretation. Varying the characteristic atomic length from 0.050 to 0.300 nm changes the normalized length ratio from approximately 33.3 to 199.8 and the normalized velocity ratio from approximately 671.7 to 274.2. The absolute

numerical values are therefore dependent on the characteristic scale adopted for the atomic system. However, the large separation of the energy and timescale ratios persists throughout the investigated range.

The nuclear-charge analysis produces a related result. The characteristic velocity follows approximately a $Z^{1/2}$ dependence, while the characteristic Coulomb energy and force scales are proportional to Z . Consequently, increasing the nuclear charge modifies the numerical comparison but does not eliminate the large separation between the atomic and planetary energy and force scales.

The mass-number analysis also reveals that the near-unity normalized mass hierarchy is not a universal property of all possible atomic systems. Under the simplified mass model used here, $\mathcal{R}_M$ changes systematically with A . The value obtained for Gold-197 is therefore a particular result of the selected nuclear and planetary parameters rather than evidence for a universal correspondence between atomic and planetary mass hierarchies.

Taken together, these findings suggest that the atomic–planetary analogy is best understood as a scale-dependent structural analogy rather than as a dynamical equivalence. The two systems share a conceptual arrangement involving a dominant central object and a smaller associated component, but their characteristic physical scales and governing interactions are substantially different. This conclusion is consistent with the historical development of atomic theory, in which planetary models provided useful conceptual and mathematical guidance but were ultimately superseded by quantum-mechanical descriptions [4, 5, 6].

The principal contribution of the present study is therefore the systematic organization of this distinction through dimensionless characteristic-scale comparisons. Rather than assigning a single qualitative verdict to the atomic–planetary analogy, the framework identifies which quantities exhibit numerical correspondence and which display extreme scale separation. The results indicate that the analogy cannot be represented by a single scale-independent transformation between atomic and planetary systems.

Several limitations must nevertheless be emphasized. The atomic model uses a simplified central Coulomb interaction and a selected characteristic atomic length. Gold is a many-electron relativistic atom, and a complete description would require an appropriate quantum-mechanical treatment. Similarly, the Sun–Earth system is treated as an idealized two-body system, whereas actual planetary dynamics involve additional bodies and perturbations. These limitations do not invalidate the scaling exercise, but they define the scope within which its results should be interpreted.

Future work can extend the framework by replacing

the single Gold-197 reference system with a systematically selected set of atomic systems and by comparing them with multiple planetary systems. Such an extension would allow the dimensionless parameter space to be explored more comprehensively and would provide a stronger test of whether any non-trivial scaling relationships persist across different physical systems.[13]

9. Limitations and Scope

The quantitative framework developed in this study is intentionally simplified. Its purpose is to compare selected characteristic physical scales between an atomic and a planetary system rather than to provide complete dynamical models of either system. The interpretation of the results must therefore remain within the assumptions defined in Sec. 2.

9.1. Limitations of the Atomic Model

The atomic reference system is represented through a central Coulomb characteristic-scale construction. This approximation does not describe the complete electronic structure of Gold-197. A real Gold-197 atom is a many-electron quantum system in which electron–electron interactions, screening, relativistic effects, and quantum-mechanical state structure contribute to the electronic properties.

Consequently, the characteristic velocity, timescale, energy, force, and angular-momentum quantities constructed in this work should not be interpreted as exact observables of a Gold-197 electron. They are model-defined scales introduced to enable a controlled dimensionless comparison.

The adopted characteristic atomic length, $r_{\text{Au}} = 0.144$ nm, is another source of model dependence. Different definitions of an atomic size or characteristic electronic length can lead to different numerical values of the dimensionless ratios. The sensitivity analysis in Sec. 6 partially quantifies this dependence.

9.2. Nuclear-Radius Approximation

The nuclear radius is represented using the empirical relation

$$R_N = r_0 A^{1/3}. \quad (92)$$

This relation provides a convenient characteristic nuclear scale, but it does not constitute a complete description of nuclear structure. The actual charge distribution and nuclear radius can depend on nuclear deformation, proton and neutron distributions, and the specific definition of the measured radius.

Therefore, the nuclear radius used in the present analysis should be understood as a reference scale rather than as a complete structural description of the Gold-197 nucleus.

9.3. Limitations of the Planetary Model

The planetary reference system is treated as an idealized Sun–Earth two-body system. In the real Solar System, planetary motion is affected by the gravitational perturbations of other bodies, non-circular orbital motion, and additional dynamical effects.

These effects are outside the scope of the present characteristic-scale comparison. The two-body approximation is adopted to provide a well-defined and reproducible planetary reference scale.

9.4. Interpretation of Energy and Force Ratios

The interaction-energy and interaction-force ratios require particular caution. The atomic quantities are constructed from a Coulomb reference scale, whereas the planetary quantities are constructed from Newtonian gravitational scales.

The resulting ratios therefore represent comparisons of the specific characteristic quantities defined in this model. They should not be interpreted as universal dimensionless measures of the fundamental relative strengths of electromagnetism and gravity.

Similarly, the angular-momentum comparison uses a classical characteristic angular momentum for the atomic reference system. It does not represent a comparison between a quantum angular-momentum eigenvalue and a planetary orbital angular momentum.

9.5. Dependence on the Normalization Scheme

The dimensionless ratios depend not only on the physical parameters but also on the characteristic quantities selected for normalization. A different physically motivated normalization scheme could therefore produce different numerical ratios.

For this reason, the present framework should be regarded as a specific comparative methodology rather than as a unique universal mapping between atomic and planetary physics. The conclusions are restricted to the characteristic scales and normalization definitions explicitly specified in this work.

9.6. Scope and Future Extensions

The framework can be extended by replacing the simplified atomic model with quantum-mechanical characteristic quantities. For example, future studies could use experimentally determined atomic radii, effective electronic length scales, quantum binding energies, or expectation values of relevant observables.

A broader comparison could also include multiple atomic species rather than Gold-197 alone. Varying atomic number, mass number, and electronic structure across a systematically selected set of atoms would allow the persistence of the observed scaling behavior to be tested over a wider parameter space.

On the planetary side, the framework could be extended to multiple planetary systems and to additional characteristic quantities such as orbital eccentricity, orbital period, angular frequency, and multi-body dynamical parameters.

Such extensions would provide a stronger test of whether any dimensionless correspondences identified in the present study are specific to the selected Gold-197–Sun–Earth comparison or represent more general patterns across physically distinct systems.

The scope of the present work is therefore deliberately limited to establishing and testing a reproducible characteristic-scale framework. The framework does not claim that atomic and planetary systems obey identical dynamics or can be transformed into one another by a universal scaling law.

10. Conclusion

This study developed a dimensionless characteristic-scale framework for quantitatively examining the long-standing analogy between atomic and planetary systems. A Gold-197 atomic reference system was compared with the Sun–Earth system using characteristic length, velocity, timescale, mass hierarchy, interaction energy, interaction force, and angular momentum.

The results demonstrate that the degree of correspondence depends strongly on the physical quantity considered. Among the quantities examined, the normalized mass hierarchy is closest to unity, with

$$R_M \approx 1.0864. \quad (93)$$

This numerical correspondence, however, is not accompanied by a similar correspondence across the remaining characteristic scales. The normalized timescale, energy, force, and angular-momentum ratios remain separated from unity by many orders of magnitude, while the length and velocity ratios also differ substantially from unity.

The sensitivity analysis further shows that the numerical values of the dimensionless ratios depend systematically on the characteristic atomic length, nuclear charge, and mass number. Nevertheless, the principal qualitative distinction between the near-unity mass hierarchy and the strongly separated dynamical and interaction scales remains evident throughout the parameter ranges explored in this study.

The central conclusion is therefore that the atomic–planetary analogy represents a limited structural and dimensional correspondence rather than a general physical or dynamical equivalence. A similarity in one dimensionless characteristic quantity cannot be taken as evidence that the underlying systems share the same dynamics, interactions, or microscopic structure.

The framework developed here provides a reproducible quantitative method for examining such cross-scale correspondences. Its principal value lies in separating numerical similarity from physical equivalence and in providing a basis for extending the comparison to other atomic systems, characteristic quantum-mechanical quantities, and planetary or astrophysical systems.

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